

Phrasing the Problem

The goal is to synthesize a spatial mechanism to build a 3 DOF platform (roll, pitch and vertical translation) with following requirements:

To ensure precision, motion should be actuated by slider pair (P) distributed on three chains. All the other joints on those three chains should comprise of turning pair (R).

To reduce load on turning pair, a redundant chain can be added to provide additional lift to the platform. The redundant chain need not have high precision.

Compliant joint has displacement limit and non-zero rotation stiffness. Free-body analysis and dimension study should be carried out to improve and verify the design.

Mechanism Synthesis

The graph is as follows (Figure 1)

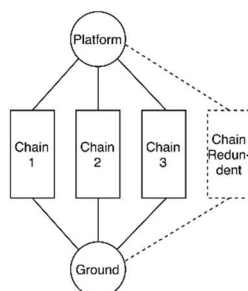


Figure 1 Mechanism graph.

We start with chain 1, 2 and 3. For simplicity, we set those chain 2 and 3 to be identical as chain 1. Combining DOF formula and previous requirements, the only candidates for chain 1 is listed in [1]

PRRRR, RPRRR, RRP RR [1]

I want to reduce link number, thus compressing turning pair (R) into universal joint (U) and ball joint (S), thus getting [2].

PSR, RPS, UPU [2]

Then, I dismissed any candidate that has ball joint (S). Making ball joint (S) out of turning pair (R) can encounter Gimbal lock; it is also far less compact than a universal joint (U). Thus, the only candidate is the traditional UPU chain.

Current Work

I am now working on constructing a kinematic model and a static model to analytically evaluate local displacement and load at each complaint joint. A CAD file is made to help model construction (Figure 2).

I do have some idea about synthesis of redundant chains, but haven't checked whether they are feasible.

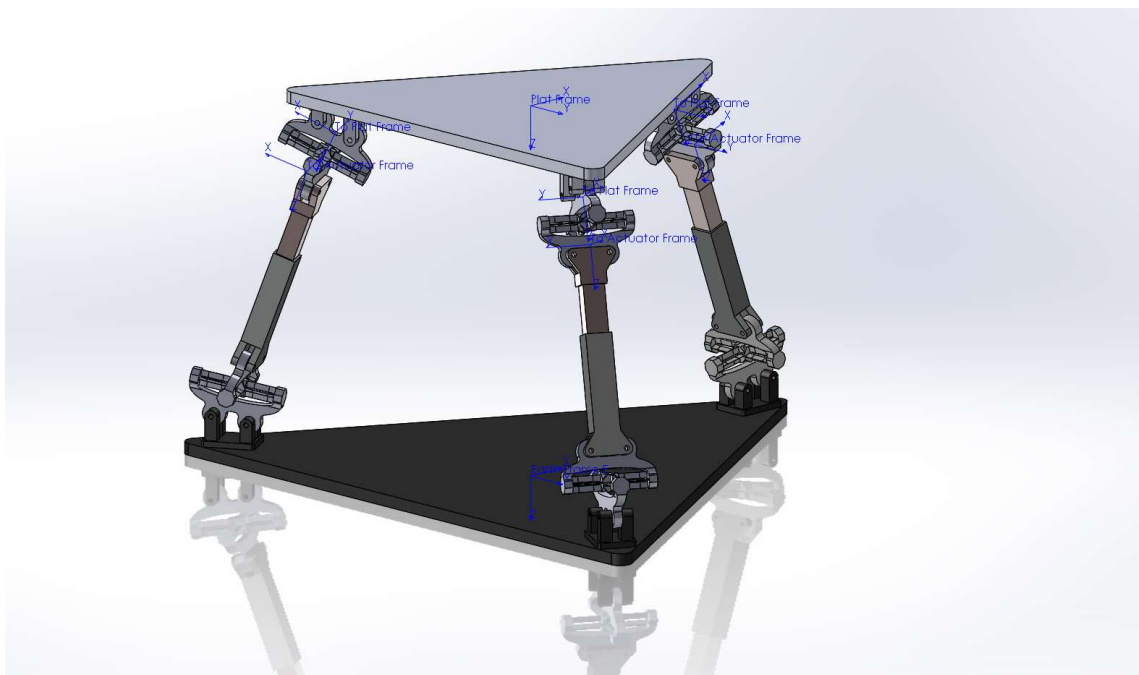


Figure 2 CAD file that helps to construct kinematic and static model